

Def.

On an interval I , the differential equation of the form

$$L(y) = a_n \cdot y^{(n)}(t) + a_{n-1} \cdot y^{(n-1)}(t) + \dots + a_1 y'(t) + a_0 y(t) = b(t)$$

is called the linear equation. If $b(t) \equiv 0$, we say that homogeneous.

Observation.

The L is a differential operator on $C^{(n)}$, because: for any $c \in \mathbb{R}$ and $y_1, y_2 \in C^{(n)}$,

$$L(cy_1 + y_2) = \sum_{i=0}^n a_i \cdot (cy_1 + y_2)^{(i)}(t) = \sum_{i=0}^n [c \cdot a_i y_1^{(i)}(t) + a_i y_2^{(i)}(t)] = c \cdot L(y_1) + L(y_2).$$

Order of 2.

Thm. Let $a_1, a_2 \in \mathbb{R}$ be constants, and consider the linear equation

$$L(y) = y'' + a_1 y' + a_2 y = 0.$$

If r_1, r_2 are distinct roots of the characteristic polynomial

$$p(r) = r^2 + a_1 r + a_2,$$

then $\phi(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ ($C_1, C_2 \in \mathbb{R}$)

is the general solution for $L(y) = 0$.

If r_1 is a repeated root of p , then

$$\phi(x) = C_1 e^{r_1 x} + C_2 x e^{r_1 x} \quad (C_1, C_2 \in \mathbb{R})$$

is the general solution.

Proof. observe that

$$L(e^{rx}) = r^2 \cdot e^{rx} + a_1 r e^{rx} + a_2 e^{rx} = (r^2 + a_1 r + a_2) \cdot e^{rx} = p(r) \cdot e^{rx}$$

and

$$L(x e^{rx}) = (x e^{rx})'' + a_1 (x e^{rx})' + a_2 (x e^{rx})$$

$$= (r e^{rx} + r e^{rx} + r^2 x e^{rx}) + a_1 e^{rx} + a_1 r x e^{rx} + a_2 x e^{rx}$$

$$= (r^2 + a_1 r + a_2) \cdot x e^{rx} + (2r + a_1) e^{rx}$$

$$= (p(r) \cdot x + p'(r)) \cdot e^{rx}$$

$p(r) = p'(r) = 0 \Leftrightarrow r$ is multiple root of p , so Proved.

Non-Homogeneous form.

Thm. If Q_1, Q_2 are two solutions of

$$y'' + a_1 y' + a_2 y = 0 \quad \text{on an interval } I \text{ containing } x_0.$$

then $W(Q_1, Q_2)(x) = e^{-a_1(x-x_0)} \cdot W(Q_1, Q_2)(x_0).$

Proof. We have $\begin{cases} Q_1'' + a_1 Q_1' + a_2 Q_1 = 0 \\ Q_2'' + a_1 Q_2' + a_2 Q_2 = 0 \end{cases}$ so multiplying $-Q_2$ and Q_1 at first and second, respectively,

$$\text{and adding: } \begin{cases} -Q_1'' Q_2 - a_1 Q_1' Q_2 - a_2 Q_1 Q_2 = 0 \\ Q_1 Q_2'' + a_1 Q_1 Q_2' + a_2 Q_1 Q_2 = 0 \end{cases}$$

$$= (Q_1 Q_2'' - Q_1'' Q_2 + a_1 (Q_1 Q_2' - Q_1' Q_2)) = 0 \quad \dots (*)$$

Notice that $(Q_1 Q_2' - Q_1' Q_2)' = Q_1 Q_2'' + Q_1' Q_2' - Q_1' Q_2' - Q_1'' Q_2 = Q_1 Q_2'' - Q_1'' Q_2$.
Therefore the (*) is: for $W = W(Q_1, Q_2)$,

$$W' + a_1 W = 0 \Rightarrow W = C \cdot e^{-a_1 x}$$

Now, $C = W(x_0) \cdot e^{a_1 x_0}$, we obtain the result. $\square$

Consider that:

$$L(y) = y'' + a_1 y' + a_2 y = b(y)$$

If ψ, ψ_p are solutions of $L(y) = b(x)$, then

$$L(\psi - \psi_p) = b - b = 0, \text{ so } \psi - \psi_p \text{ is a solution of } L(y) = 0.$$

Therefore, $\psi - \psi_p = C_1 Q_1 + C_2 Q_2$ where Q_1, Q_2 are solutions of $L(y) = 0$.
So, if we have one 'particular' solution ψ_p , then we get the general solution.

Particular solution.

Let $\mathcal{Y}_p(x) = u_1(x) \cdot \mathcal{Q}_1(x) + u_2(x) \cdot \mathcal{Q}_2(x)$. Then,

$$b = \mathcal{Y}_p'' + a_1 \mathcal{Y}_p' + a_2 \mathcal{Y}_p$$

$$\begin{aligned} &= u_1'' \mathcal{Q}_1 + 2u_1' \mathcal{Q}_1' + u_1 \mathcal{Q}_1'' + a_1 u_1' \mathcal{Q}_1 + a_1 u_1 \mathcal{Q}_1' + a_2 u_1 \mathcal{Q}_1 \\ &+ u_2'' \mathcal{Q}_2 + 2u_2' \mathcal{Q}_2' + u_2 \mathcal{Q}_2'' + a_1 u_2' \mathcal{Q}_2 + a_1 u_2 \mathcal{Q}_2' + a_2 u_2 \mathcal{Q}_2 \\ &= \underbrace{u_1 \cdot L(\mathcal{Q}_1) + u_2 \cdot L(\mathcal{Q}_2)}_{=0} \end{aligned}$$

$$+ (u_1'' \mathcal{Q}_1 + u_2'' \mathcal{Q}_2) + 2(u_1' \mathcal{Q}_1' + u_2' \mathcal{Q}_2') + a_1(u_1' \mathcal{Q}_1 + u_2' \mathcal{Q}_2)$$

Now, if $u_1' \mathcal{Q}_1 + u_2' \mathcal{Q}_2 = 0$, then $u_1'' \mathcal{Q}_1 + u_2'' \mathcal{Q}_2 + u_1' \mathcal{Q}_1' + u_2' \mathcal{Q}_2' = 0$,

So, only $u_1' \mathcal{Q}_1' + u_2' \mathcal{Q}_2' = b$ is remained.

Let's solve the equation $\begin{cases} u_1' \mathcal{Q}_1 + u_2' \mathcal{Q}_2 = 0 \\ u_1' \mathcal{Q}_1' + u_2' \mathcal{Q}_2' = b \end{cases}$

Multiplying $\mathcal{Q}_2'$ to $u_1' \mathcal{Q}_1 + u_2' \mathcal{Q}_2 = 0$, then we obtain

$$u_1' \mathcal{Q}_1 \mathcal{Q}_2' + u_2' \mathcal{Q}_2 \mathcal{Q}_2' = 0. \text{ Substituting } u_2' \mathcal{Q}_2' = b - u_1' \mathcal{Q}_1'$$

$$\begin{aligned} \text{Then we obtain } u_1' \mathcal{Q}_1 \mathcal{Q}_2' + \mathcal{Q}_2 b - u_1' \mathcal{Q}_1' \mathcal{Q}_2 \\ = u_1' \cdot W(\mathcal{Q}_1, \mathcal{Q}_2) = -\mathcal{Q}_2 b. \end{aligned}$$

Therefore, proceed to similar process to u_2 , we obtain:

$$u_1' = \frac{-\mathcal{Q}_2 b}{W(\mathcal{Q}_1, \mathcal{Q}_2)} \quad \text{and} \quad u_2' = \frac{\mathcal{Q}_1 b}{W(\mathcal{Q}_1, \mathcal{Q}_2)}.$$

Consequently,

$$\mathcal{Y}_p(x) = \int_{x_0}^x b(t) \cdot \begin{vmatrix} \mathcal{Q}_1(t) & \mathcal{Q}_2(t) \\ \mathcal{Q}_2'(t) & \mathcal{Q}_1'(t) \end{vmatrix} \cdot \frac{1}{W(\mathcal{Q}_1, \mathcal{Q}_2)(t)} dt.$$

